PROJECT PLANNING & MANAGEMENT

Stock Market Analytics & Investment Decision Support Dashboard

DEPI — Professional Data Analysis Track
Phase 1 Documentation

Project Information	Details
Team Members	Member 1: [أحمد عبدالحميد محمد الجندي] Member 2: [أحمد عبدالباسط مصطفى] Member 3: [رقية محمد عبدالسلام شلبي] Member 4: [زينب عبدالمنعم دياب] Member 5: [بيشوي سامي يوسف بقطر] Member 6: [محمد إبراهيم عبدالله غنيم]
Project Supervisor / Instructor	[ENG: Adel Youssef]
Submission Date	16 October 2026
Final Project Deadline	30 November 2026

1. Project Proposal

1.1 Project Name

Stock Market Analytics & Investment Decision Support Dashboard

1.2 Problem Statement

Investors and analysts need a clear and consistent way to monitor stock market performance, compare companies, and understand changes in stock prices over time. Raw stock data can be difficult to interpret without structured analysis and clear visualizations. This project aims to transform historical and periodically refreshed stock data into useful insights about price movements, returns, trading activity, and risk.

1.3 Project Objectives

- Collect daily stock market data for a selected group of companies from Yahoo Finance.
- Organize and prepare the data for reliable analysis.
- Analyze opening, closing, highest, and lowest prices, as well as trading volume when available.
- Calculate daily returns and compare company performance across selected periods.
- Evaluate historical price volatility and other relevant risk indicators.
- Build an interactive Power BI dashboard with company and date filters.
- Rank or group companies according to transparent, predefined performance criteria to support informed decisions, without guaranteeing future returns.

1.4 Dataset Description and Source

The planned dataset will contain daily stock market observations for selected publicly traded companies. The expected fields include Date, Open, High, Low, Close, and Volume, depending on availability for the selected ticker symbols and market. Yahoo Finance is the intended source. The team will verify the availability, update frequency, coverage, and permitted use of the data before implementation.

The final list of companies, historical date range, market coverage, dataset size, and exact data acquisition method will be documented after the team confirms them.

1.5 Tools and Technologies

Tool	Planned Use
Python	Automating data retrieval or preparation and calculating analytical measures, subject to the team's chosen approach.
SQL	Structured storage, querying, filtering, and analysis if included in the final workflow.
Power BI	Interactive dashboard, KPI cards, filters, and visual comparisons.

2. Project Plan & Gantt Chart

The following is a proposed schedule leading to the final project deadline of 30 November 2026.

#	Task / Phase	Start Date	End Date	Duration
1	Confirm project scope, companies, dataset fields, and business questions	13 Oct 2026	16 Oct 2026	4 days
2	Acquire sample data and inspect data quality	17 Oct 2026	21 Oct 2026	5 days
3	Clean, transform, and structure the data	22 Oct 2026	28 Oct 2026	7 days
4	Perform exploratory analysis and calculate KPIs	29 Oct 2026	5 Nov 2026	8 days
5	Build the data model and develop dashboard visuals	6 Nov 2026	15 Nov 2026	10 days
6	Validate calculations, filters, and refresh process	16 Nov 2026	21 Nov 2026	6 days
7	Finalize documentation, insights, and presentation	22 Nov 2026	29 Nov 2026	8 days
8	Final review and submission	30 Nov 2026	30 Nov 2026	1 day

3. Task Assignment & Roles

Team Member	Primary Role	Main Responsibilities
رقية محمد عبدالسلام شلبي	Python Developer	Collecting stock market data from Yahoo Finance, retrieving data, and preparing it for analysis. Contributing to project documentation and presentation.
أحمد عبدالباسط مصطفى	Python Developer	Data cleaning, exploratory data analysis (EDA), statistical analysis, and extracting insights. Contributing to project documentation and presentation.
زينب عبدالمنعم دياب	SQL Developer & GitHub Manger	Writing SQL queries, managing the GitHub repository, organizing project files, and uploading team deliverables. Contributing to project documentation and presentation.
أحمد عبدالحميد محمد الجندي	SQL Developer	Database design, creating tables, writing SQL queries, and analyzing stock market data. Contributing to project documentation and presentation.
بيشوي سامي يوسف بقطر	Power Bi Developer	Designing interactive dashboards, creating KPIs, visualizations, and filters, and presenting analytical insights. Contributing to project documentation and presentation.
محمد إبراهيم عبدالله غنيم	Power Bi Developer	Developing dashboard pages, improving visualizations, and supporting interactive analysis and reporting. Contributing to project documentation and presentation.

4. Risk Assessment & Mitigation Plan

Risk	Potential Impact	Mitigation Plan
Data source or access limitations	Some fields or update frequencies may not be available through the chosen method.	Test access early, document source limitations, and agree on a feasible refresh schedule before building the dashboard.
Missing, duplicate, or inconsistent observations	Incorrect records may distort returns and comparisons.	Profile the dataset, define quality rules, document cleaning decisions, and validate results after transformation.
Market holidays or delayed updates	The latest available date may differ from the current calendar date.	Use the latest available timestamp/date, account for market schedules, and display the last successful update.
Incorrect calculations or comparisons	Returns, rankings, or risk indicators may be misleading.	Validate formulas against sample records, independently check selected calculations, and test filters and aggregations.
Scope growth and time constraints	Too many companies, indicators, or refresh requirements may delay delivery.	Prioritize essential requirements, limit the initial company list, assign owners, and review progress against the Gantt schedule.

5. Key Performance Indicators (KPIs)

#	KPI	Definition	Why It Matters	Calculation
1	Opening Price	The first traded price recorded for the selected daily interval, as provided by the source.	Shows the stock's starting price for the interval.	Use the Open field for the selected date.
2	Closing Price	The last recorded price for the selected daily interval, as provided by the source.	Supports daily comparisons and historical trend analysis.	Use the Close field; document whether adjusted prices are used in return calculations.
3	Daily Return (%)	Percentage change in closing price between consecutive trading observations.	Compares relative price performance across stocks with different price levels.	((Current Close / Previous Close) - 1) × 100.
4	Daily Price Range	Difference between the day's highest and lowest prices.	Describes the intraday price range.	High - Low.
5	Trading Volume	Number of shares traded during the selected interval, as reported by the source.	Helps assess trading activity and liquidity context.	Use the Volume field; sum or compare it at the chosen reporting level.
6	Average Trading Volume	Average reported trading volume over a selected period.	Provides context for comparing activity across time.	Average of daily Volume values over the selected period.
7	Historical Volatility	A measure of how much daily returns vary over a selected period.	Helps compare historical price variability, not future risk with certainty.	Standard deviation of daily returns over the selected period; define the window and annualization convention if used.
8	Period Return (%)	Percentage change between the first and last selected closing-price observations.	Summarizes historical performance over a chosen period.	((Last Close / First Close) - 1) × 100.

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